Global Knowledge®

Angular Fundamentals

Inleiding

Peter Kassenaar –
info@kassenaar.com

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Peter Kassenaar

- Trainer, author, developer – since 1996
- Specialty: *"Everything JavaScript"*
- JavaScript, ES6, Angular, NodeJS, TypeScript, React, Vue, Phonegap

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Sr. frontend developer ProductIP, Ede, NL

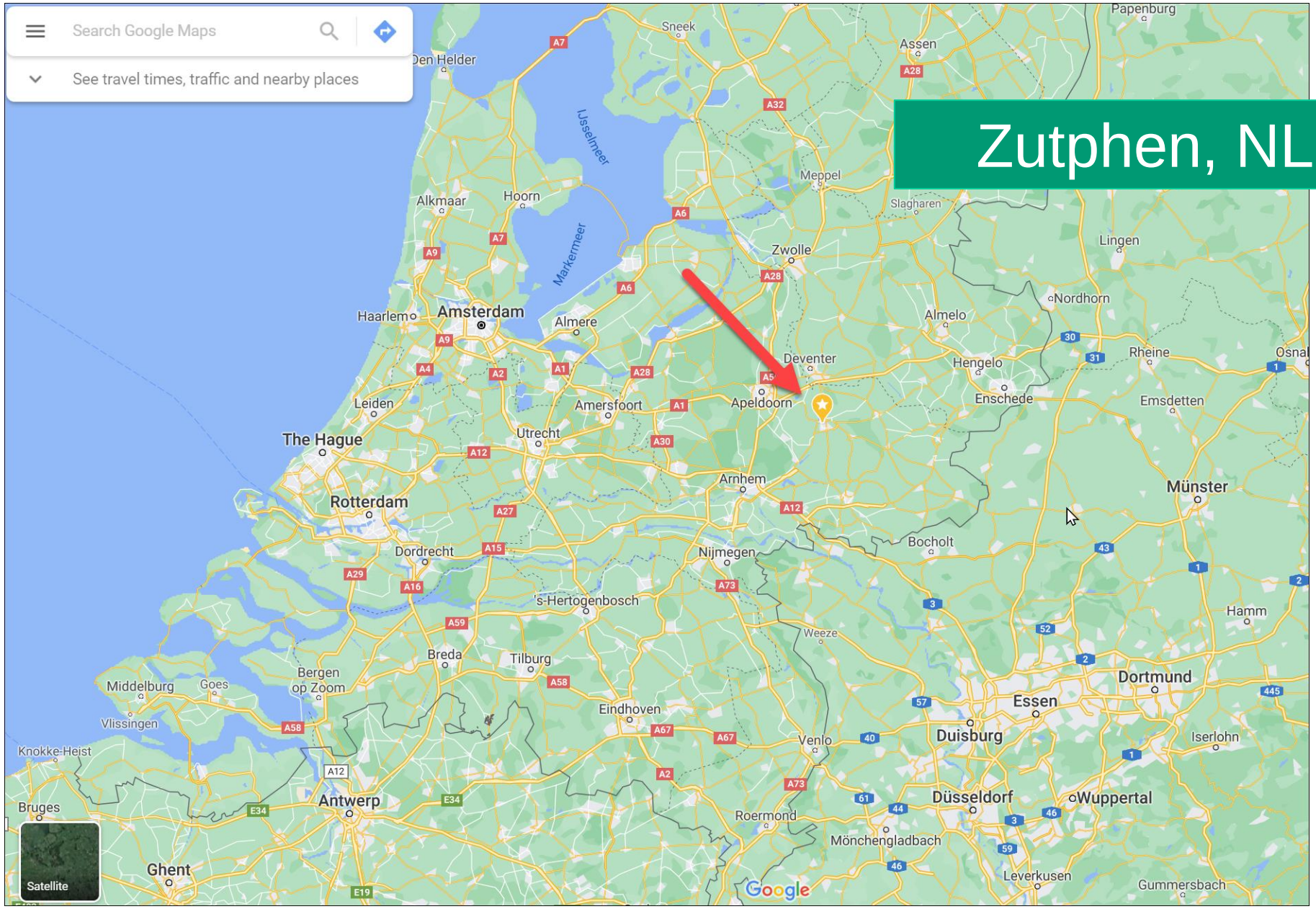
www.productip.com

@PeterKassenaar

PeterKassenaar

pkas06

Zutphen, NL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www.angulartraining.nl

github.com/PeterKassenaar/ilionx

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PeterKassenaar chore: updated README.md cbdb375 · 2 minutes ago 6 Commits

Angular	chore: updated README.md	2 minutes ago
JavaScript	chore: updated structure	4 minutes ago
.gitignore	Added slides and updated README.md	3 weeks ago
LICENSE	Initial commit	3 weeks ago
README.md	chore: updated README.md	2 minutes ago

README | MIT license

ilionx

Slides and sample code on various trainings Ilionx, Young Professional Program, Fall 2024

- See the different folders in the repo for slides, example code and more. This repo will be deleted on Nov. 1, 2024. Make sure you have a copy before that date.

Questions? Feel free to reach out at info@kassenaar.com.

About

Slides and sample code on the training JavaScript Fundamentals, Ilionx, Fall 2024

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Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

Octotree

Over jullie

Stel jezelf kort voor

Voorkennis webdevelopment, (mobile/web-) apps?

Voorkennis andere (web)talen?

Verwachtingen van de cursus?

Concrete projecten?

Specifieke vragen of technieken die je wilt behandelen?

Agenda – globaal – 3 dagen

30 sept, 1, 2 oktober 2024 – ma. – wo.

~ 9:00 start

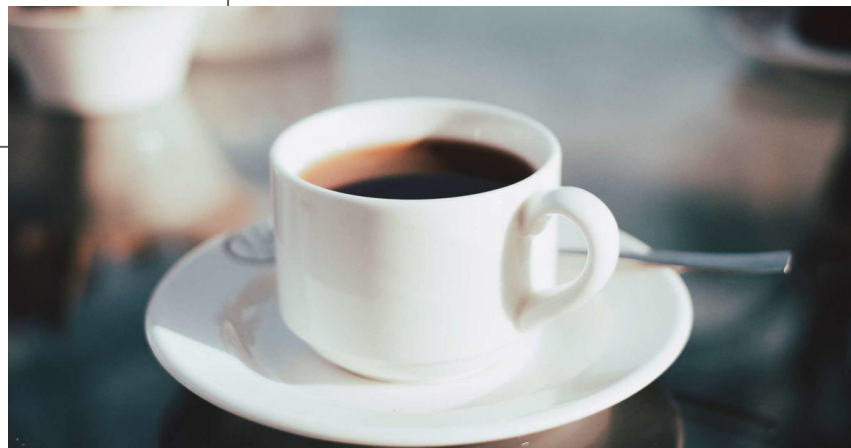
~ 10:00, 11:00 short break

~ 12:00 lunch

~ 12:45 middagsessie

~ 13:00, 14:00 short break

~16:00 Einde



Woensdag: iets eerder stoppen

Doel van de training

*Je wordt **geen** Angular wizard in 3 dagen (sorry)*

maar….

Doelen

1. Je leert over de structuur en architectuur van Angular Apps. Van een kleine `hello-world` app tot een grote Enterprise applicatie.

2. Je bent bekend met de belangrijkste *Angular concepten* van het framework. Specifieke details kun je altijd Googelen.

3. Je hebt enige hands-on ervaring met het maken van apps en componenten, services, API's/backends, component communicatie.

4. Je hebt een algemeen begrip van de manier waarop moderne web apps worden gemaakt met Angular, TypeScript en build tools.

Agenda - Fundamentals

- Introductie & geschiedenis - waarom Angular, TypeScript?
- Kernbegrippen in Angular 2 – 17
- CLI, Hello World in Angular – inzicht in boilerplate-code
- Angular in depth (modules):
  - Components
  - ECMAScript 2015 + TypeScript
  - Data binding
  - Dependency Injection (DI) – more components
  - Services en http, Observables (RxJS), communicatie backend
  - Intercomponent communication

Materialen

Software	(Angular, NodeJS + NPM, editor, browser)
Handouts	(Github - PDF)
Oefeningen	(Github)
Websites	(online)

angular.dev/

Vandaag

- Dag 1 – Intro & Data binding
  - Theorie - Introductie & geschiedenis – waarom Angular
  - Angular CLI, TypeScript
  - Hello World in Angular –boilerplate-code
    - Components + Modules
  - Concepts, context & architecture
  - Data binding

2 Richtlijnen

1. Oefeningen
   - Maar: neem ook vooral zijpaden, experimenteer, lees verder, maak een eigen project, app, website…

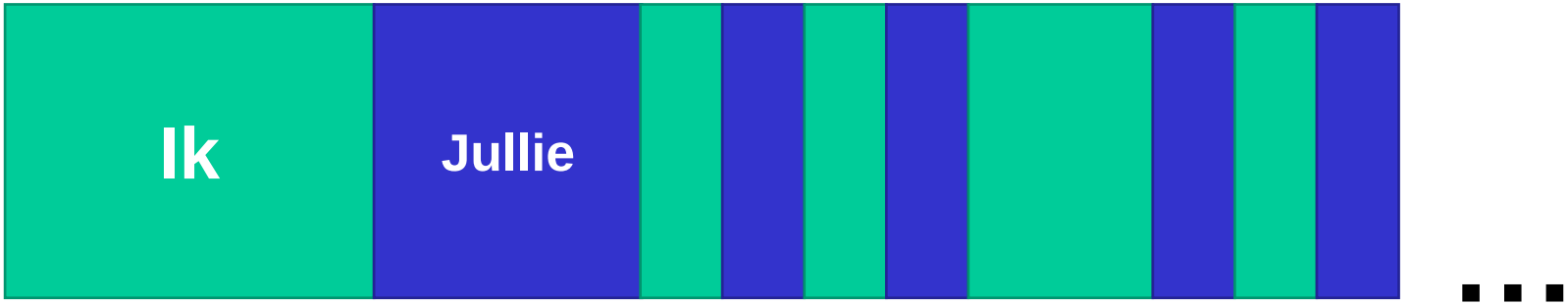
2. Voorbeeldcode
   - Als ondersteuning bij de oefeningen, zie boven
   - Work in progress – check de Angular-site!
   - github.com/PeterKassenaar/voorbeeldenAngular2

Advanced warning - First morning

Not so much code…

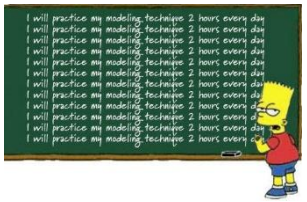
Concepts, architecture, structure

Globale werkwijze



On Workshops…

- … Zien er uit zoals deze slide 👇
- Worden uitgevoerd tussen de theoretische delen
- Zijn NIET regel-voor-regel uitgeschreven. Zelf nadenken.
- De tijd gedurende de training is misschien te kort om alle workshops af te ronden
  - Doe ze in je eigen tijd
  - Je weet dan tenminste de concepten waar de workshop over gaat
  - Keuze – er is meer te bespreken, ik geef dat voorrang
- De voorbeeldcode bevat vaak (maar niet altijd!)De ‘oplossing’ van de workshop. Gebruik dit! Daar zijn ze voor
  - Maar het zou natuurlijk mooi zijn als je eigen data gebruikt



Vragen?

Angular vs. The Rest

Differences, similarities, new features

Addressing the "WHY" question!

WHY, would we want to use a frontend framework.

It is all HTML, CSS and JavaScript right?

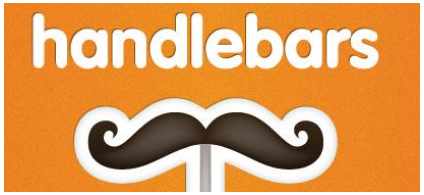
Rethorical question:

speed, consistency, not re-inventing the wheel, community, performance, testing….

"Do we want to go back to the jQuery days?"

Old school web apps

HTML + templates



Data Binding

Routing

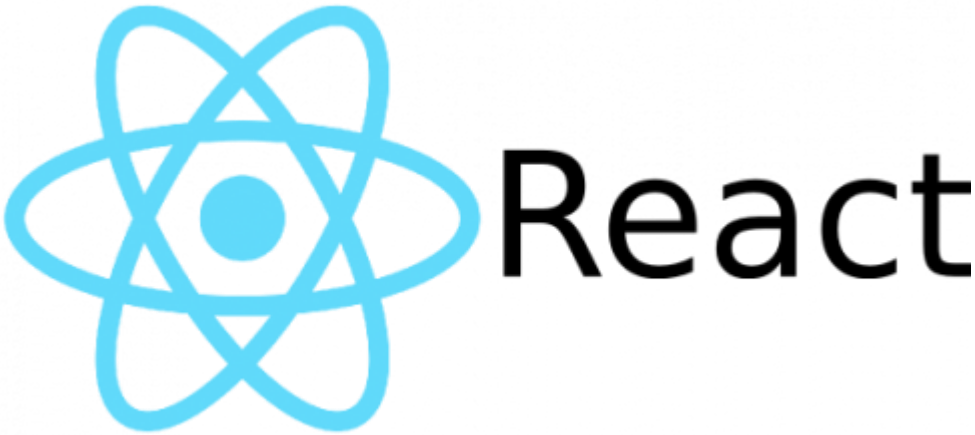
DOM-manipulation

Mobile development

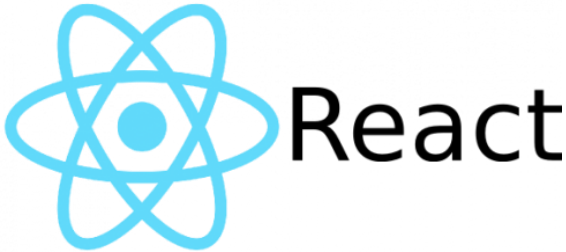
...

“The Frankenstein Framework”

Front-end Frameworks – the big four



Similarities



- Creating Single Page Applications
- Based on components
- Data binding, props, events, routing, state management, …
- Huge ecosystem
- Huge community
- High adaptation rate

Differences (apart from syntax)

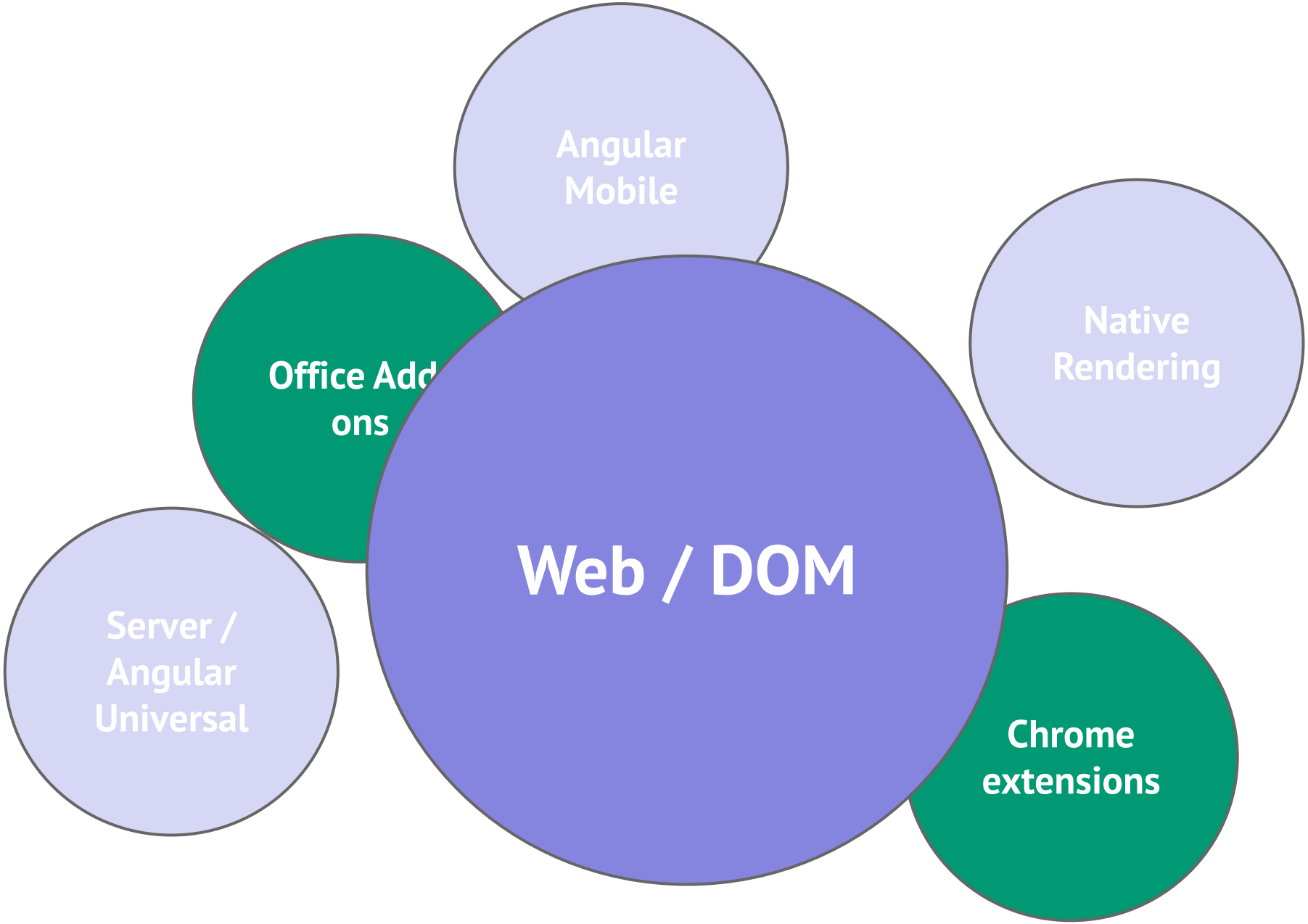
- Point of departure: HTML template, enhanced with framework specific tags and attributes
- One-stop-shop / solution

- Point of departure: JavaScript, JavaScript, JavaScript (JSX)
- Build-all-yourself / choice anxiety

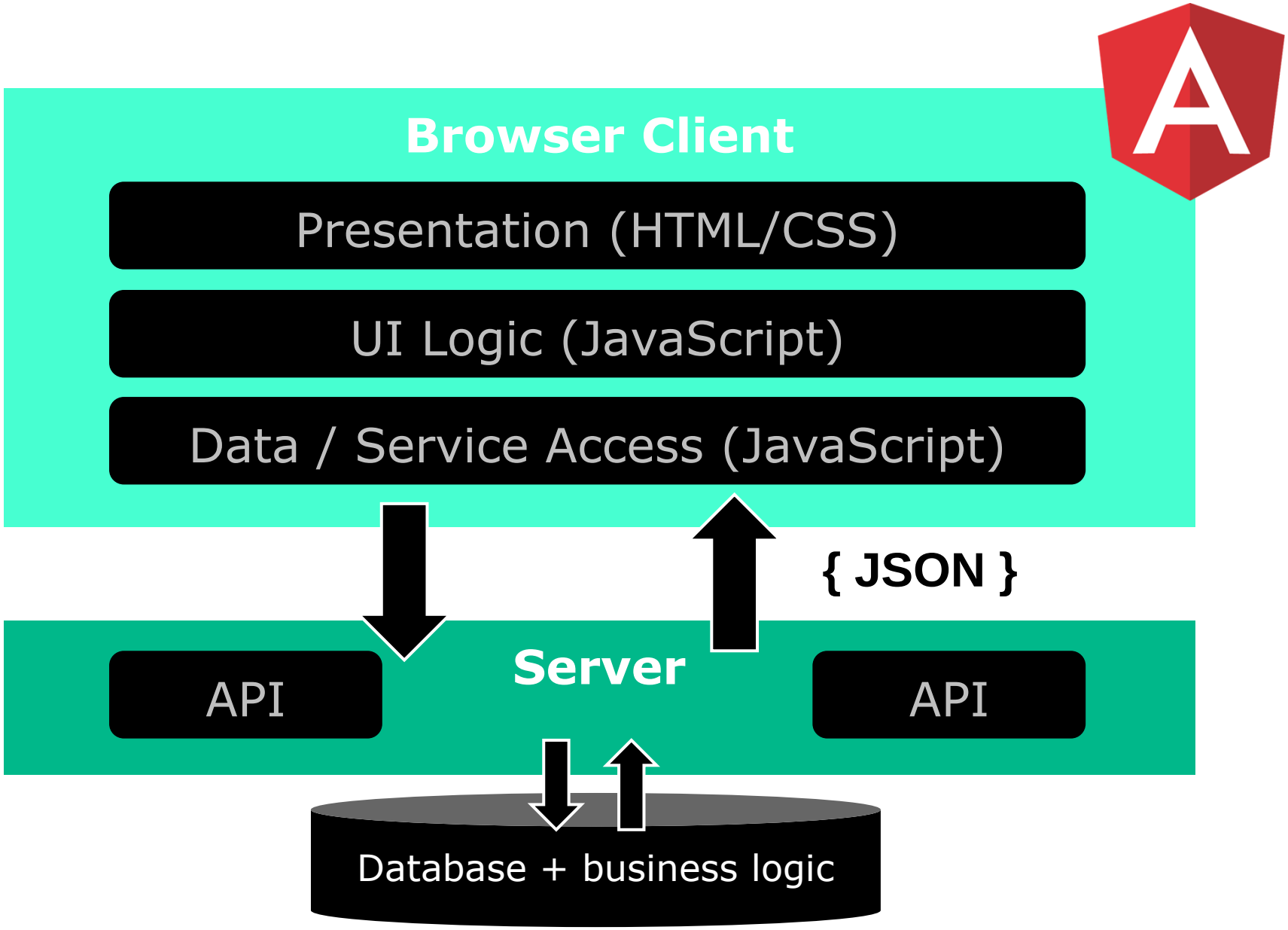
Platform

Platform Features

	Scaffolding	Code completion & Refactoring	Debugging
Tooling	Angular CLI	Language Services	DevTools
Libraries	Material 2	Mobile	Universal
	AOT- Compile	Change Detection	Renderer
Core	Components & Dependency Injection	Decorators	Zones


Angular Mobile
Office Add ons
Native Rendering
Web / DOM
Server / Angular Universal
Chrome extensions

Single Page Application



@IgorMinar, Nov. 9, 2018

Angular updates

- Angular 2 – 14 **Grotendeels 'meer van hetzelfde'**
  - Soms zijn er breaking changes
  - Zelf checken: https://update.angular.io/
  - Updates van verwante libraries (RxJS, Angular Material)
- Belangrijke milestones
  - Angular 2 – breekt met structuur en architectuur van Angular JS (1.x)
  - Angular 6 – Introductie modern Angular CLI
  - Angular 9 – Ivy Compiler
  - Angular 14 – Standalone components
  - Angular 19 – Current - Fall 2024

angular.dev/update-guide

Update Guide

Select the options that match your update

Angular versions

From v. 17.0 To v. 18.0

Application complexity

✓ Basic | Medium | Advanced

Shows information for all Angular developers.

Other dependencies

- [ ] I use ngUpgrade to combine AngularJS & Angular
- [ ] I use Angular Material
- [x] I use Windows

Show me how to update!

Angular Versies en -Long Time Support

→ https://angular.io/guide/releases

Secure | https://angular.io/guide/releases

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Style Guide
Glossary
API
stable (v6.0.4)

- 6 months of active support, during which regularly-scheduled updates and patches are released, as described above in Release frequency.
- 12 months of long-term support (LTS). During the LTS period, only critical fixes and security patches will be released.

The following table provides the support status and key dates for Angular version 4.0.0 and higher.

Version	Status	Release Date	LTS Start Date	LTS End Date
^4.0.0	LTS	March 23, 2017	September 23, 2017	September 23, 2018
^5.0.0	LTS	November 1, 2017	May 1, 2018	May 1, 2019
^6.0.0	Active	May 3, 2018	November 3, 2018	November 3, 2019

Deprecation practices

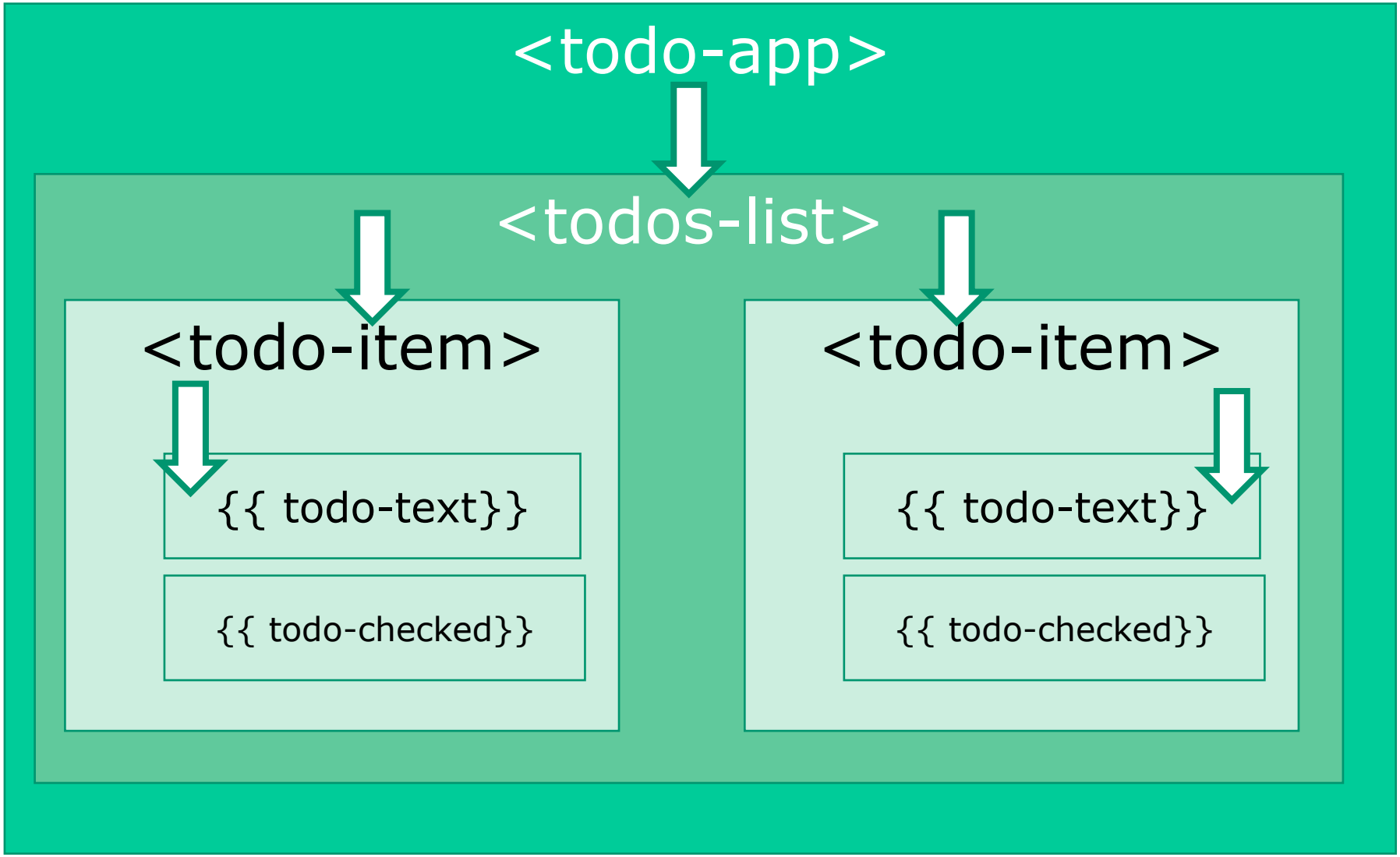
Sometimes "breaking changes", such as the removal of support for select APIs and features, are necessary to innovate and stay current with new best practices, changing dependencies, or changes in the (web) platform itself.

To make these transitions as easy as possible, we make two commitments to you:

- We work hard to minimize the number of breaking changes and to provide migration tools when possible.
- We follow the deprecation policy described here, so you have time to update your apps to the latest APIs and best practices.

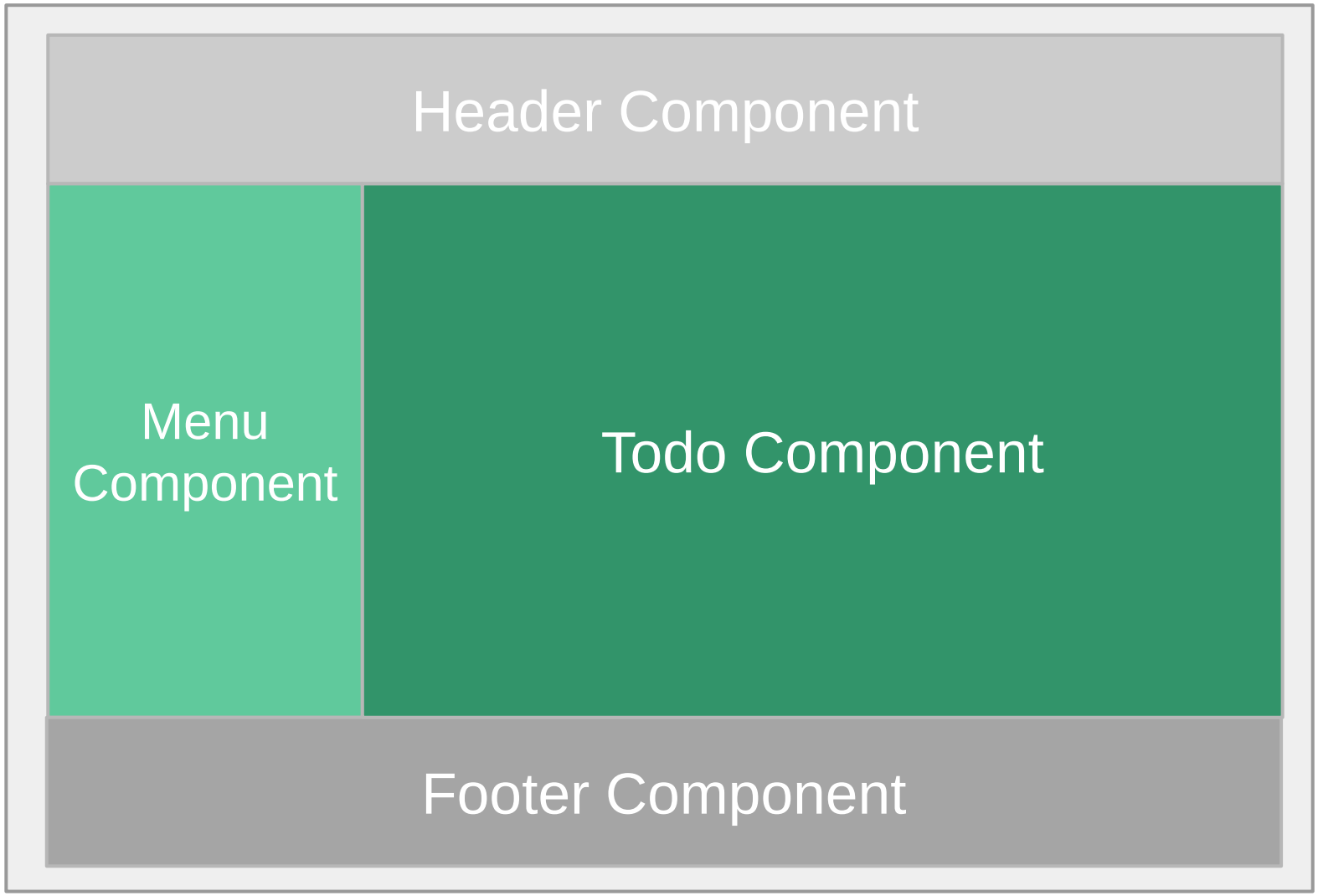
Angular versioning and releases
Angular versioning
Release frequency
Release schedule
Support policy
Deprecation practices
Public API surface
Angular Labs

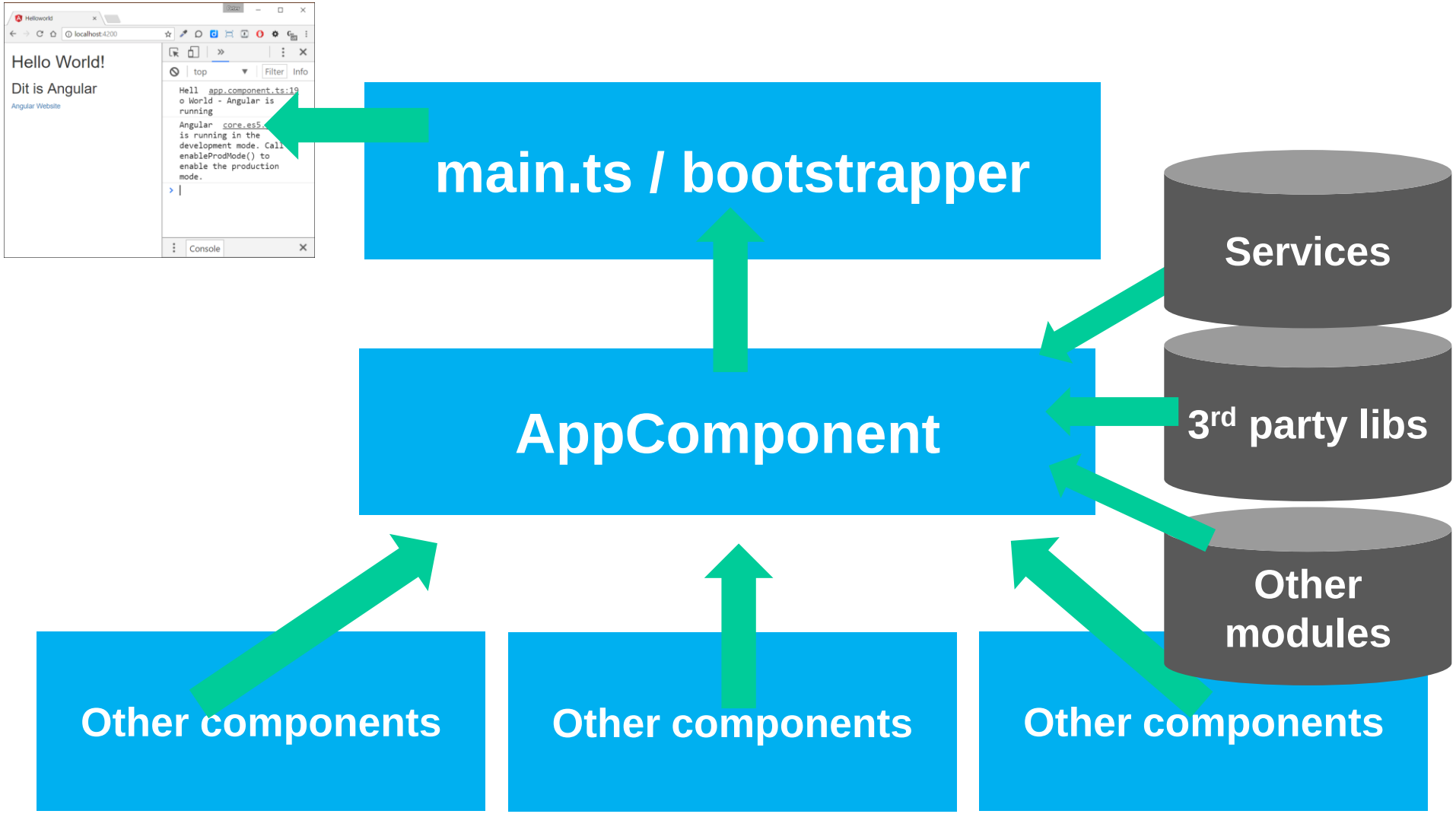
Angular – components



“An Angular-app is a tree of components”

Components – visual representation




Helloworld
localhost:4200
Hello World!
Dit is Angular
Angular Website
top
Filter
Info
Hell app.component.ts:19
o World - Angular is
running
Angular core.es5.
is running in the
development mode. Call
enableProdMode() to
enable the production
mode.
Console
main.ts / bootstrapper
Services
AppComponent
3rd party libs
Other modules
Other components
Other components
Other components

Let’s write some code

Hello World in Angular

Angular development dependency: NodeJS 18.x +

Node.js v22 is now available!

node.js Learn About Download Blog Docs Certification Start typing... Ctrl K

Survey Help us shape the next 10 years of Node.js →

Run JavaScript Everywhere

Node.js® is a free, open-source, cross-platform JavaScript runtime environment that lets developers create servers, web apps, command line tools and scripts.

Download Node.js (LTS)

Downloads Node.js **v20.13.1**[1] with long-term support.
Node.js can also be installed via package managers.

Want new features sooner? Get **Node.js v22.1.0**[1] instead.

Create an HTTP Server | Write Tests | Read and Hash a F

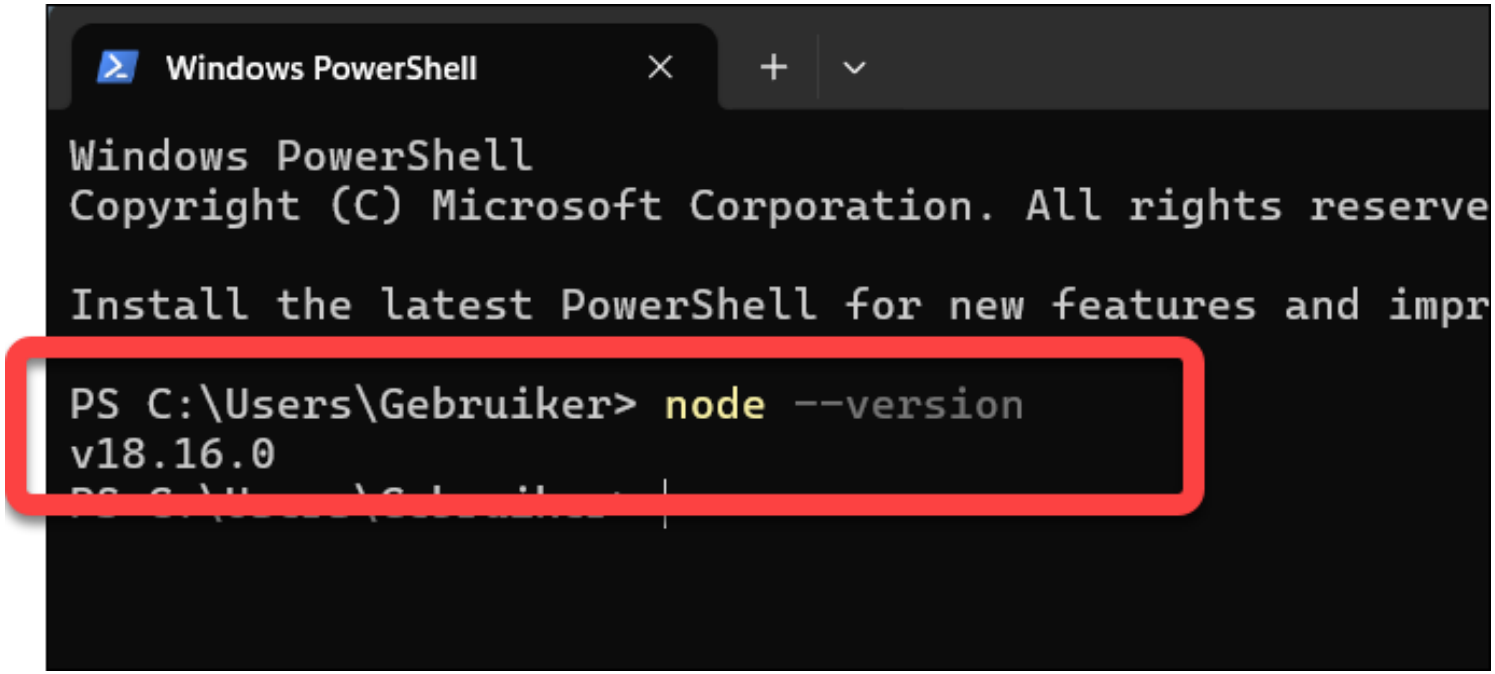
```
// server.mjs
import { createServer } from 'node:http'

const server = createServer((req, res) =
  res.writeHead(200, { 'Content-Type': '
  res.end('Hello World!\n');
});

// starts a simple http server locally o
server.listen(3000, '127.0.0.1', () => {
  console.log('Listening on 127.0.0.1:30
});

// run with `node server.mjs`
```

JavaScript | Copy to clipboard

Learn more what Node.js is able to offer with our Learning materials.

https://nodejs.org/en

Node – check your version

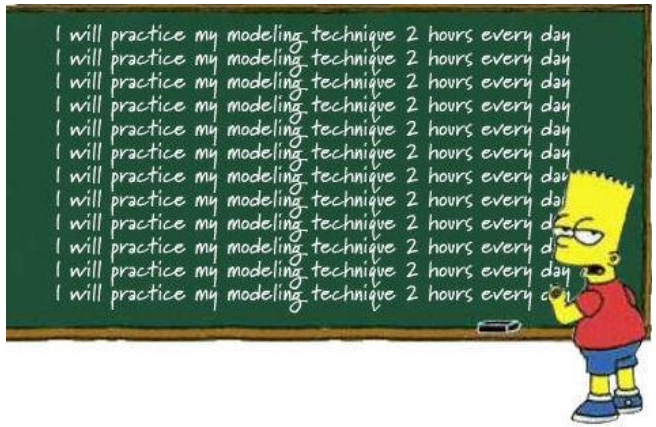


Mini workshop

- Download or clone https://github.com/PeterKassenaar/voorbeeldenAngular2

```
cd examples

cd 100-helloworld

npm install

npm start
```

- Go to browser: http://localhost:4200



localhost:4200

Hello World!

This is Angular

Angular Website

Project

voorbeeldenAngular2 C:\Users\Peter Kassenaar\Desktop\voo

- 100-helloworld
  - node_modules library root
  - src
    - app
      - app.component.ts
      - app.module.ts
    - assets
    - environments
    - favicon.ico
    - index.html
    - main.ts
    - polyfills.ts
    - styles.css
    - tsconfig.app.json
  - .angulardoc.json
  - .gitignore
  - angular.json
  - package.json
  - package-lock.json
  - tsconfig.json
  - yarn.lock

app.component.ts

```
1  import {Component, OnInit} from
2
3  @Component({
4      // 1. add component descrip
5      selector: 'hello-world',
6      template: `
7          <h1>Hello World!</h1>
8          <h2>This is Angular</h2
9          <a href="http://angular
10     `
11 })
12
13 export class AppComponent imple
14     // optional: add constructo
15     constructor() {
16     }
```

Alternative workshop

- Download and install Angular CLI
  `npm install -g @angular/cli`

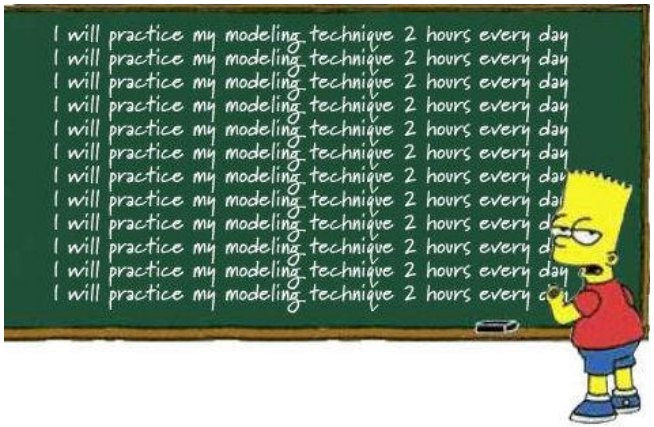
`ng new`

`Answer questions / take defaults`

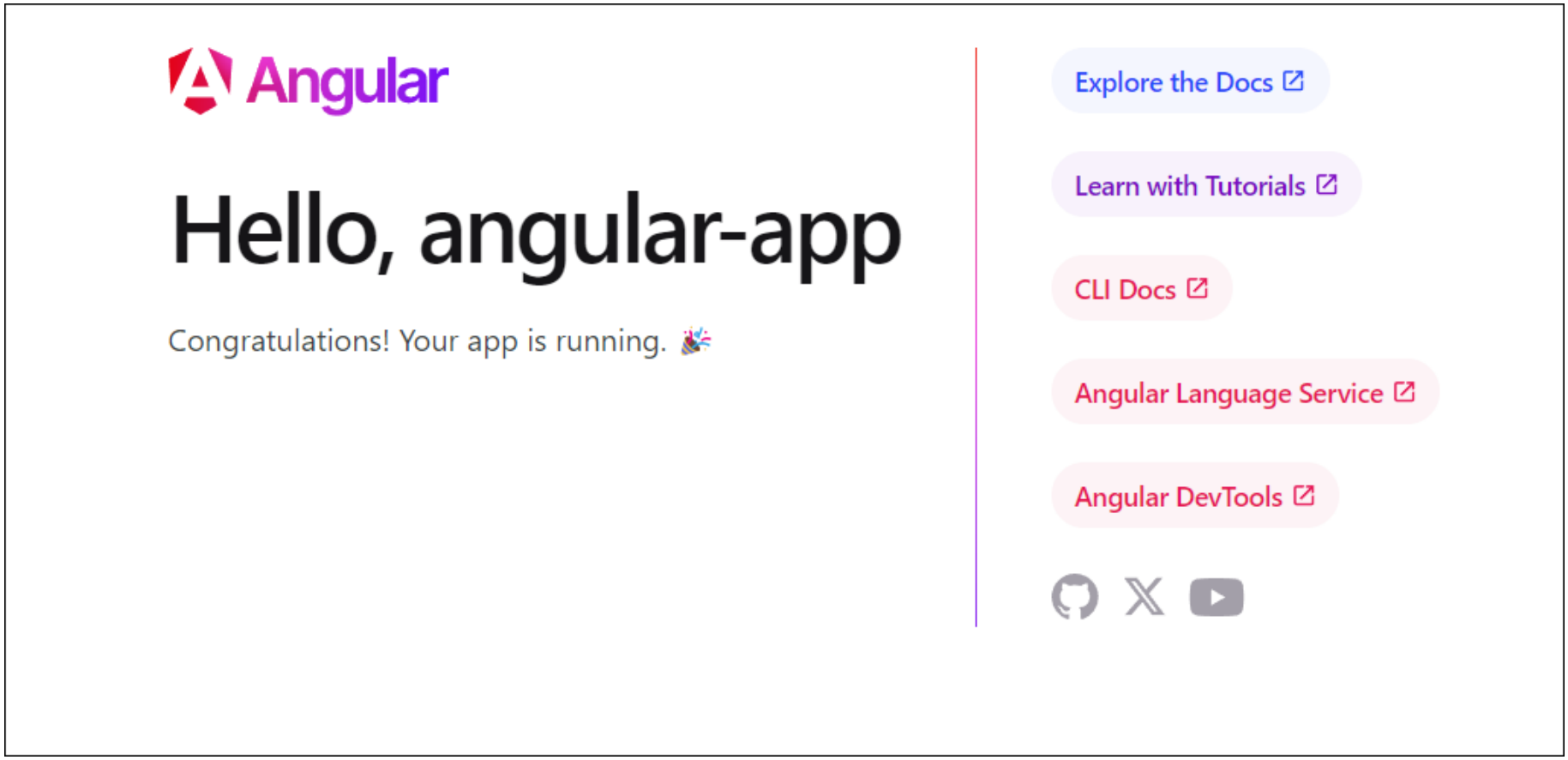
`cd-to-directory`

`npm start`

- Go to browser: http://localhost:4200



Result default angular app w/ CLI



Structure of Angular apps

Important files that are present in *all* applications

Boilerplate code for Hello World

Steps

1. Set up environment, boilerplate & libraries
   - Important configuration files
2. Angular Component(s)
3. Angular Module(s): @ngModule() ➔ NOT required in V16+
4. Bootstrap our module
5. Write HTML-pagina (`index.html`)

Boilerplate files #1/3 - `package.json`

```
{
  "name": "hello-angular",
  "description": "Voorbeeldproject bij de training Angular (C) - info@kassenaar.com",
  "version": "0.0.1",
  "license": "MIT",
  "scripts": {
    "ng": "ng",
    "start": "ng serve",
    "build": "ng build",
  },
  "private": true,
  "dependencies": {
    "@angular/animations": "16.0.0",
    "@angular/common": "16.0.0",
    "@angular/compiler": "16.0.0",
    "@angular/core": "16.0.0",
    "@angular/forms": "16.0.0",
    "rxjs": "^7,8.0",
    "zone.js": "^0.14.3"
  },
  "devDependencies": {
    "@angular-devkit/build-angular": "~0.6.0",
    "@angular/cli": "6.3.7",
    "typescript": "~5.4.2"
  },
  "author": "Peter Kassenaar <info@kassenaar.com>"
}
```

Boilerplate files #2/3 - **tsconfig.json**

```
{
  "compileOnSave"  : false,
  "compilerOptions": {
   "outDir"                : "./dist/out-tsc",
   "baseUrl"               : "src",
   "sourceMap"             : true,
   "declaration"           : false,
   "moduleResolution"      : "node",
   "emitDecoratorMetadata" : true,
   "experimentalDecorators": true,
   "target"                : "es5",
   "typeRoots"             : [
     "node_modules/@types"
   ],
   "lib"                   : [
     "es2022",
     "dom"
   ]
  }
}
```

Boilerplate files #3/3 – **angular.json**

```
{
  "$schema": "./node_modules/@angular/cli/lib/config/schema.json",
  "version": 1,
  "newProjectRoot": "projects",
  "projects": {
    "helloworld": {
      "root": "",
      "sourceRoot": "src",
      "projectType": "application",
      "architect": {
        "build": {
          "builder": "@angular-devkit/build-angular:browser",
          "options": {
            "outputPath": "dist",
            "index": "src/index.html",
            "main": "src/main.ts",
            "tsConfig": "src/tsconfig.app.json",
      …
}
```

Step 2 – Component

Convention - components in directory `/src/app`

Or: edit in `angular.json`

Filename: `src/app/app.component.ts`

```
import {Component} from ‘@angular/core';
@Component({
   selector: 'hello-world',
   template: '<h1>Hello Angular</h1>'
})
export class AppComponent {


}
```

Step 3 – @ngModule

Convention - filename: `/src/app.module.ts`

```
// Angular Modules
import {NgModule}      from '@angular/core';
import {BrowserModule} from '@angular/platform-browser';

// Custom Components
import {AppComponent} from './app.component';

// Module declaration
@NgModule({
   imports     : [BrowserModule],
   declarations: [AppComponent],
   bootstrap   : [AppComponent]
})
export class AppModule {
}
```

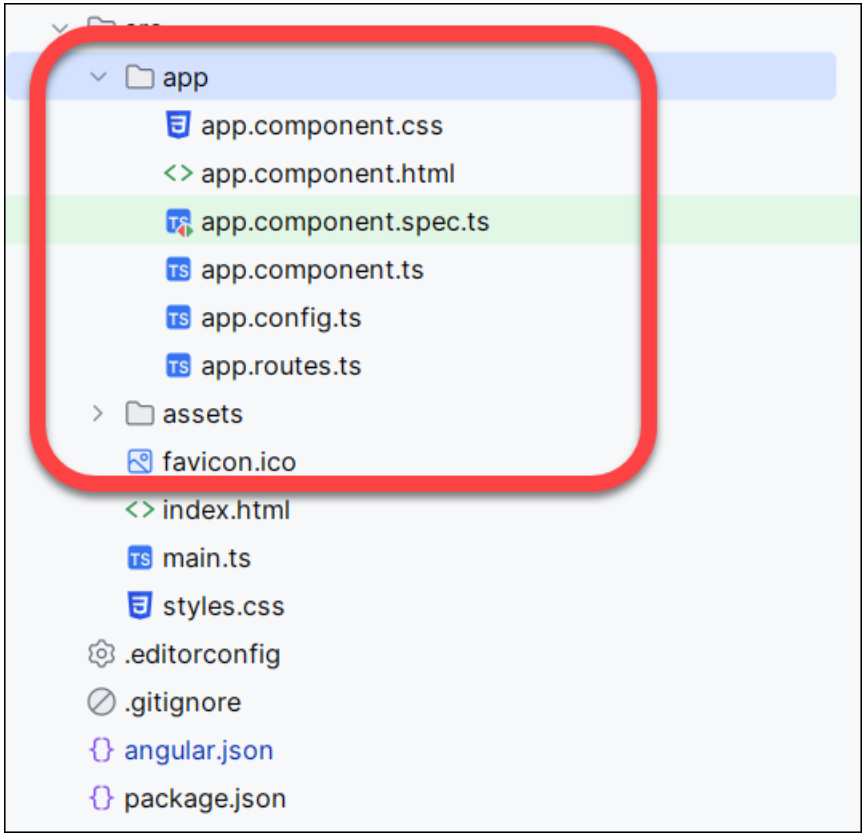
Root Module of the application

NOTE: NOT mandatory in newer applications

- As of Angular V16/17, the default choice is NOT to have Angular Modules anymore
- All applications are now composed out of standalone components

However, you still need to know about modules, as likely all existing applications use them!

New Angular structure



Some background info on Root Module

https://johnpapa.net/introducing-angular-modules-root-module/

Step 4 - bootstrap component

Best practice: bootstrap app in separate component

Convention: `main.ts`, of `app.main.ts`.

```
import {enableProdMode} from '@angular/core';
import {platformBrowserDynamic} from '@angular/platform-browser-dynamic';

import {AppModule} from './app/app.module';
import {environment} from './environments/environment';

if (environment.production) {
   enableProdMode();
}

platformBrowserDynamic().bootstrapModule(AppModule);
```

Step 5 – index.html

`index.html` - simple HTML file - expanded at runtime by WebPack

Header:

```
<html>


<head>
  <meta charset="utf-8">
  <title>Helloworld</title>
  <base href="/">


  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="icon" type="image/x-icon" href="favicon.ico">
</head>
```

Body van index.html

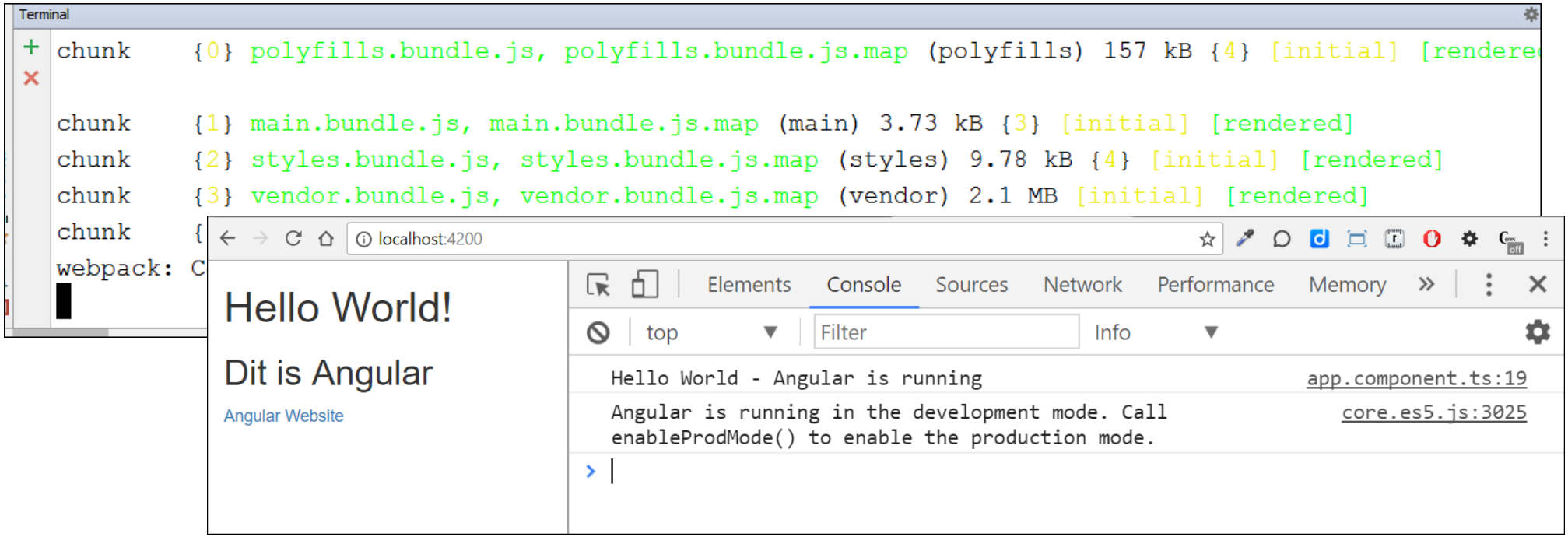
Verwijzing naar de root-component:

```
<body>
  <hello-world>
    Bezig met laden...
  </hello-world>
</body>
```

App draaien

`npm start` – draait de scriptopdracht start uit `package.json`.

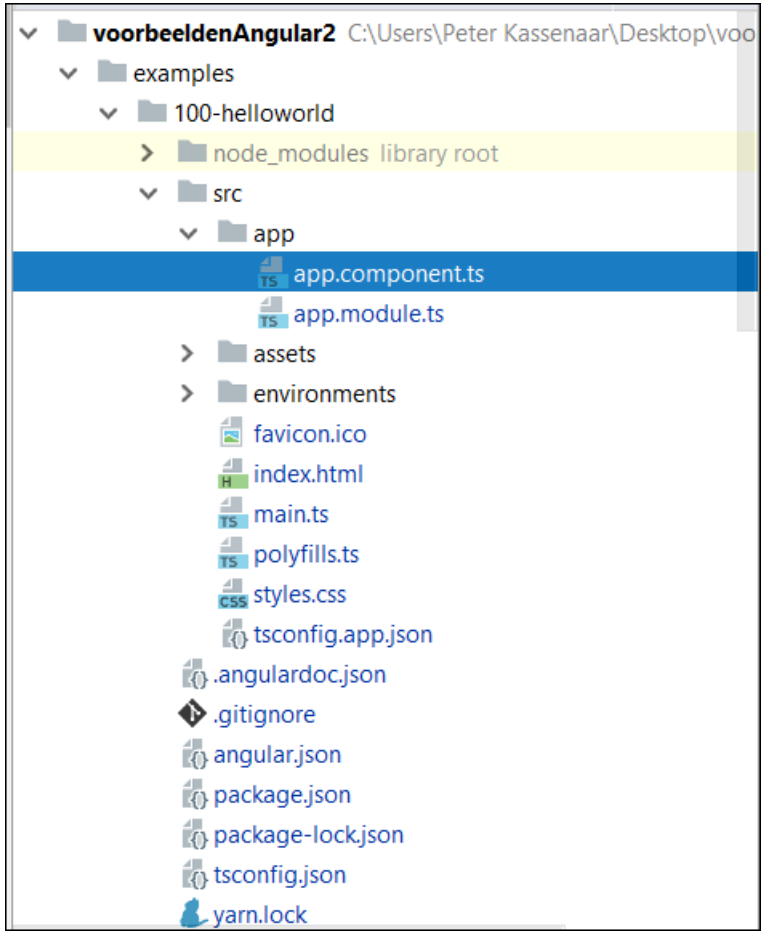
`ng serve` - start *globale* angular-cli instantie



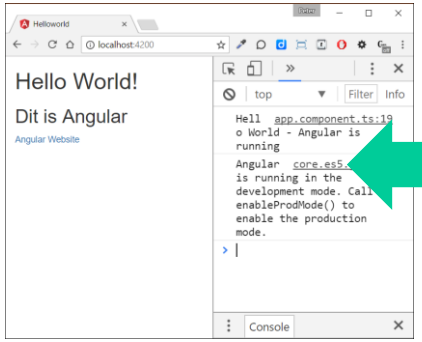
Daarna: wijzigingen aanbrengen in `app.component.ts`

– worden opgepikt door Live Reload

Basic Project Structure



Existing Angular apps



main.ts / bootstrapper

ngModule / root module

AppComponent

Services

3rd party modules

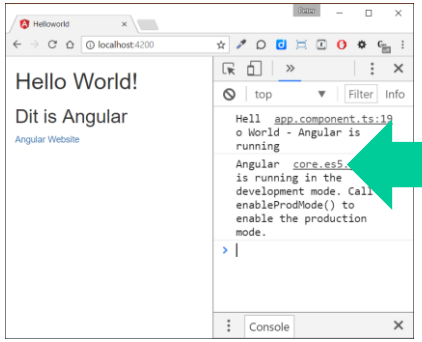
Other modules

Other components

Other components

Other components

Modern Angular apps



main.ts / bootstrapper

Services

AppComponent

3rd party libs

Other modules

Other components

Other components

Other components

(Note: no Module anymore, as this is now handled by standalone components)

Assets

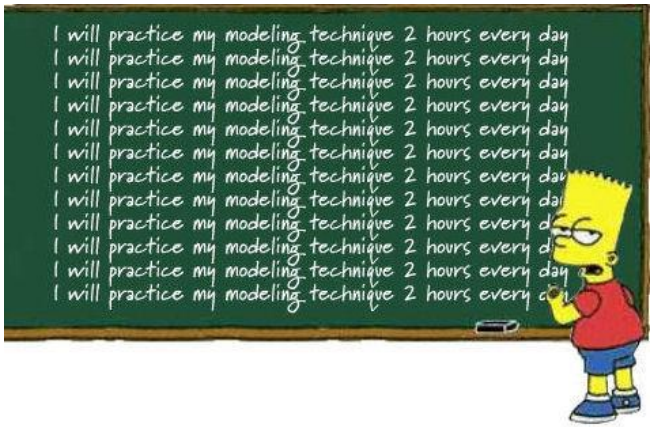
github.com/PeterKassenaar/voorbeeldenAngular2

Oefeningen en meer voorbeeldcode

Checkpoint

- Er is aardig wat boilerplate code nodig om een Angular-app te starten
- Vier stappen
  1. Set up environment, boilerplate & libraries
  2. Schrijf Angular Root Component voor de app
  3. Bootstrap de component
  4. Schrijf HTML-pagina (`index.html`)
- Daarna: app gaan uitbreiden
- Oefening `1a), 1b), 1c), 1d)`

Workshop….



Angular CLI

Snel nieuwe projecten instellen via de command line

Angular-CLI to the rescue

- Het *is* mogelijk nieuwe Angular-projecten from scratch te starten.
- Met de CLI is eenvoudiger.
- CLI-options:
  - Scaffolding
  - Generating
  - Testing
  - Building
  - AOT-Compiling
  - …

Scaffolding - Angular CLI

Projecten, componenten, routes en meer definiëren vanaf de command line

https://angular.dev/cli

Developer Tools > Angular CLI

The Angular CLI

The Angular CLI is a command-line interface tool which allows you to scaffold, develop, test, deploy, and maintain Angular applications directly from a command shell.

Angular CLI is published on npm as the `@angular/cli` package and includes a binary named `ng`. Commands invoking `ng` are using the Angular CLI.

Try Angular without local setup

If you are new to Angular, you might want to start with Try it now!, which introduces the essentials of Angular in the context of a ready-made basic online store app for you to examine and modify. This standalone tutorial takes advantage of the interactive StackBlitz environment for online development. You don't need to set up your local environment until you're ready.

Getting Started

Install Angular CLI to create and build your first app.

Get Started

Command Reference

Discover CLI commands to make you more productive with Angular.

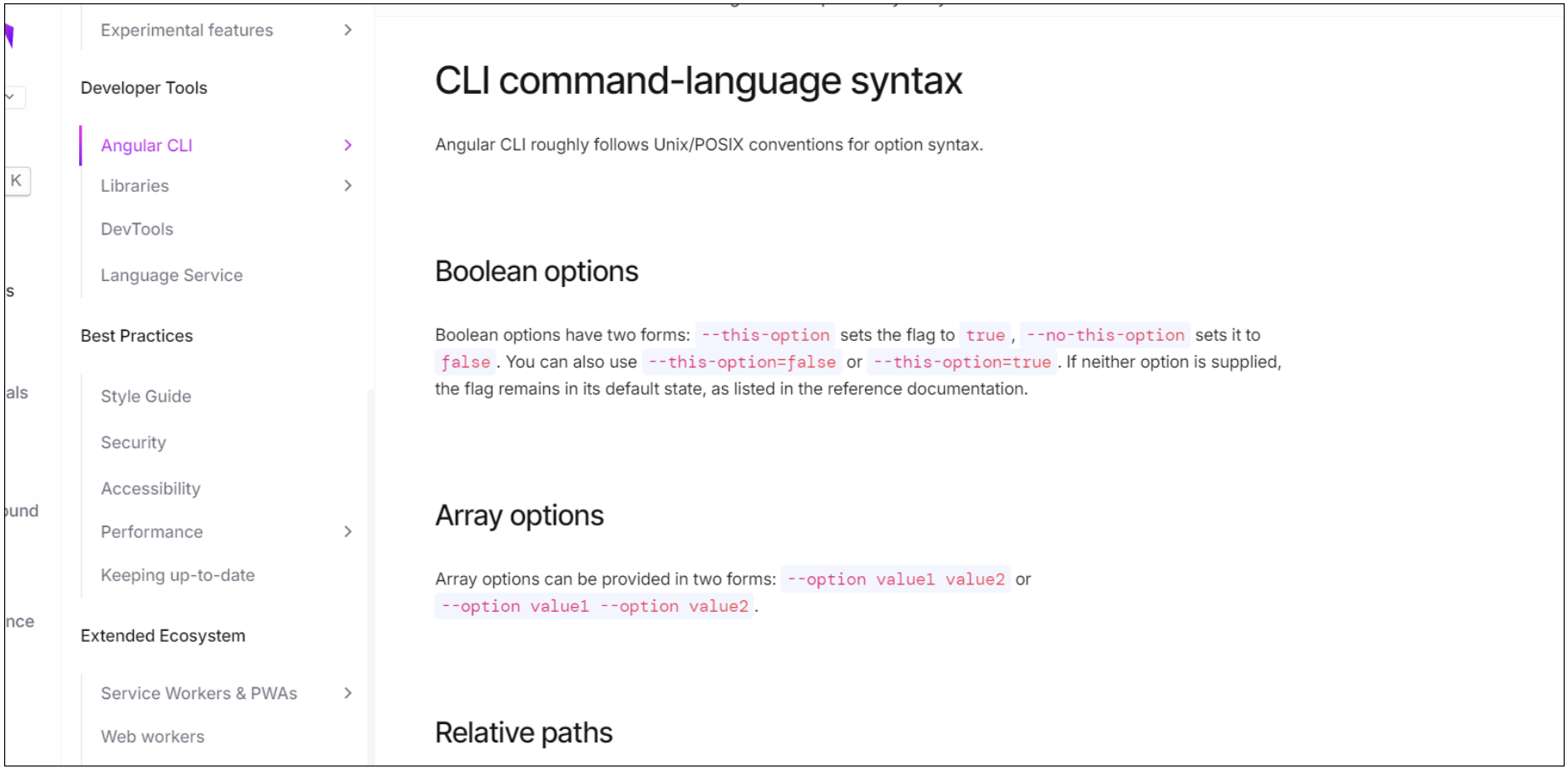
Learn More

```
npm install -g @angular/cli
```

Achtergrondinformatie – Mike Brocchi

https://www.youtube.com/watch?v=wHZe6gGI5RY

Documentatie - in de Angular Docs



angular.dev/cli

Angular 2 Code - Backend

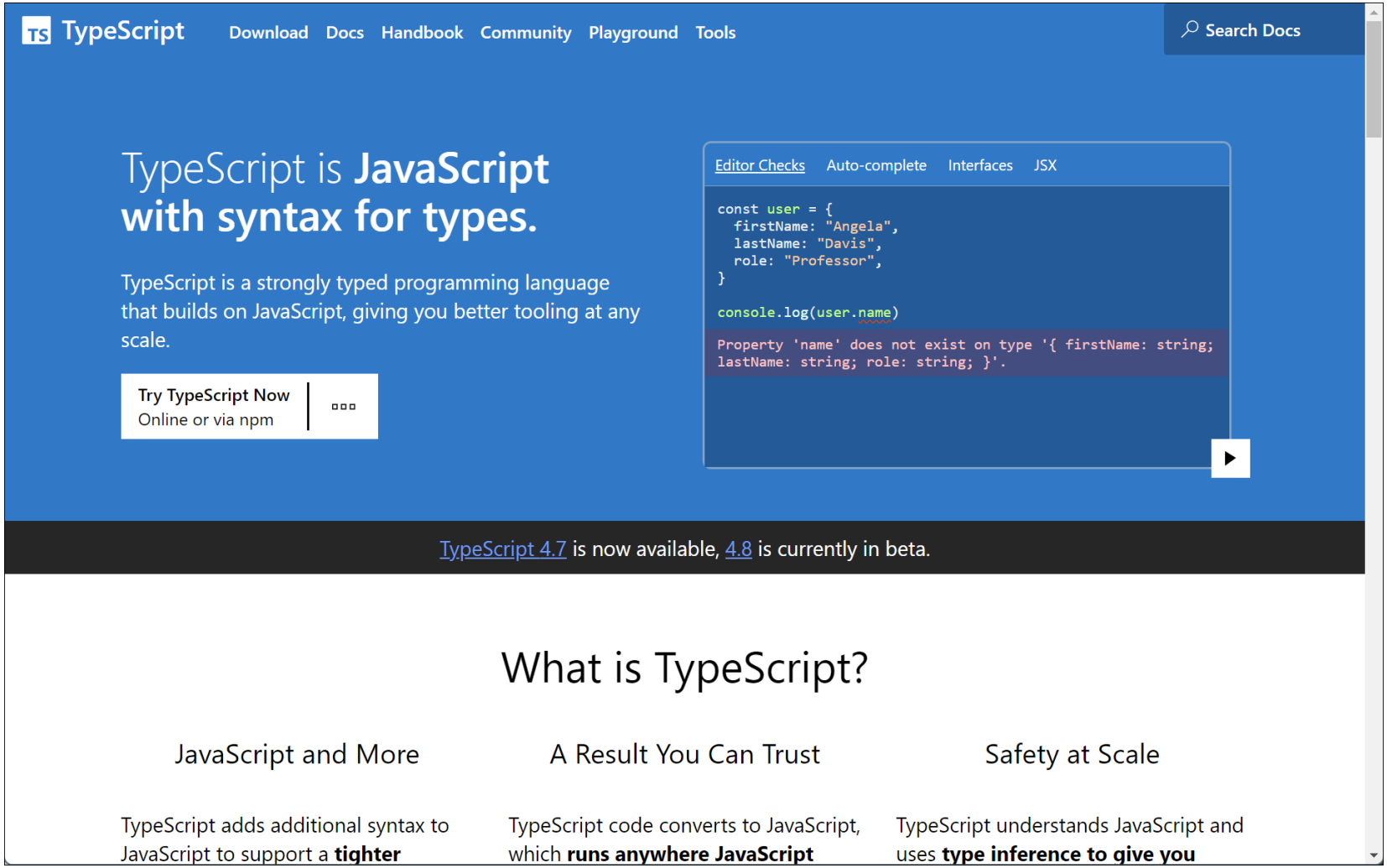
Kort over TypeScript en ES6

Programmeertalen

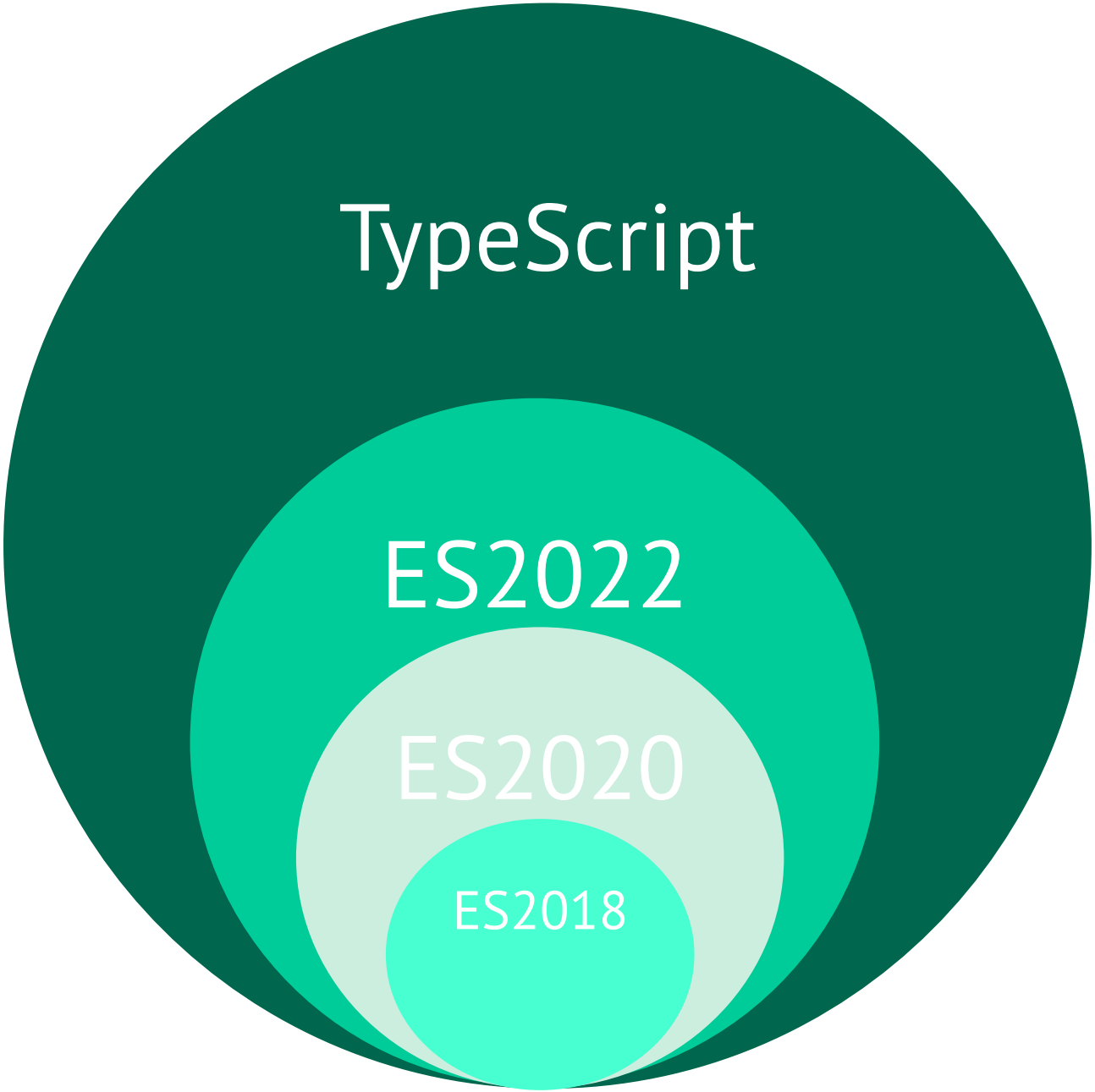
TypeScript

- Beperkt in deze module
- Voor zover nodig om Angular te gebruiken.
- Maar, TypeScript is niet beperkt tot Angular!
- Algemene superset van JavaScript
- Compileert plain JavaScript, dat browsers begrijpen
- Twee taken
    1. Compiling/Transpiling
    2. Type Safety

TypeScript website



www.typescriptlang.org


TypeScript
ES2022
ES2020
ES2018

ES6 en TypeScript

‘Modern JavaScript’, sinds ES6/ES2015

Major update van JavaScript als programmeertaal

Modules, classes en meer

Verplicht in Angular, optioneel in JavaScript, Vue, React, …

TypeScript breidt ES6 verder uit

Annotaties & types

Interfaces

Compiler,

…

TypeScript – tooling support

Types, Autocompletion.

Compile-time checking in editors.

Alles is optioneel. Je kunt altijd nog gewoon JavaScript gebruiken.

Onderdelen van een Component Class

imports

```
import { Component } from '@angular/core';
import { DataService } from './services/data-service';
```

annotations

```
@Component({
  selector: 'orders',
  templateUrl: 'orders-component.html',
})
```

class

```
export class OrdersComponent {
          …
}
```

Checkpoint

- Angular is een totaal ander framework dan AngularJS
- Component-based vs. Page-based
- Nieuwe syntaxis
- Nieuwe programmeertalen en andere nieuwe kenmerken
- Concepten komen – grotendeels – overeen
- Veel boilerplate-code nodig voor een Quickstart
- Daarna: niet meer naar omkijken. Concentreren op de componenten